

Certified Analytical Bounds for Multi-Phase Hypersonic Reachable Sets

Ethan Knox

Independent Researcher, Champaign, IL, USA

2026

Abstract

Keywords: reachable sets; occupation measures; Liouville equation; concentration-compactness; profile decomposition; comparison principle; barrier certificates; hybrid systems; hypersonic entry; entry corridor; boost-glide

MSC 2020: 34A38

Contents

1	Introduction	5
1.1	Grid-based Hamilton-Jacobi-Isaacs evaluations	5
1.2	Fixed-grid pseudospectral methods	5
1.3	Thesis statement	5
1.4	Outline of contributions	5
2	The Model Problem	6
2.1	Abstract formulation of the multi-phase optimal control problem	6
2.2	The mode alphabet	6
2.3	Trajectory families as admissible words	6
2.4	The entry corridor: hypersonic path constraints	6
2.5	Certified envelopes for constitutive and coupling effects	6
2.6	Phase boundaries: physical, constitutive, and numerical	6
2.7	Asymptotic regimes and the dimensionless groups	6
2.8	Quantity of interest and uncertainty	6
3	Infinite-Dimensional Embedding and Measure Theory	7
3.1	Sobolev space formulation	7
3.2	Occupation measures and the weak Liouville identity	7
3.3	Per-mode measures, guard measures, and linkage	7
3.4	The embedding as a linear problem on a Banach space	7
4	Concentration-Compactness and Profile Decomposition	8
4.1	Prokhorov tightness as a preliminary	8
4.2	The singular limit and the scaling group	8
4.3	Tightness of the rescaled family	8
4.4	The dichotomy	8
4.5	The skeleton: concentration on the equilibrium glide manifold	8
4.6	The bubbles: boundary-layer profiles at each skip	8
4.7	The compact residual	8
4.8	Consequence: dimension reduction for the certificate	8
5	Bounding Limits and Certificates	9
5.1	Relaxation and the direction of inclusion	9
5.2	The comparison principle	9
5.3	Constructing the certificate from the decomposition	9
5.4	Verification over the corridor	9
5.5	Exact arithmetic and machine-checkable certification	9
5.6	Inner point cloud and the bounding gap	9
5.7	Limits of the analytical construction	9
6	Verification and Results	10
6.1	Validating the degenerate limit	10
6.2	Tightness against the extremal cloud	10
6.3	Independent numerical cross-check	10
6.4	The corridor constraints as hypotheses	10
6.5	Multi-mode words	10

6.6	Numerical evidence for the profile decomposition	10
6.7	Computational cost	10
6.8	Reproducibility environment	10
7	Discussion	11
7.1	Limitations	11
7.2	Scope: the boost-glide and aeroassisted class	11
7.3	Extensions	11
A	Six-Degree-of-Freedom Equations of Motion	13
A.1	State vector and frames	15
A.2	Translational equations	15
A.3	Rotational equations	15
A.4	Aerodynamic forces and moments	15
A.5	Atmosphere	15
A.6	Per-mode forms	15
A.7	Boundary discipline	15
B	Corridor Constraints and Certified Envelopes	15
B.1	The corridor constraints in polynomial form	15
B.2	Compactness of the corridor	15
B.3	Aerodynamic coefficient envelopes	15
B.4	Ablation and aerothermoelastic envelopes	15
B.5	How the envelopes enter the certificate	15
C	Polynomial Embedding of the Entry Dynamics	15
C.1	Augmentation	15
C.2	The augmented state and its constraint set	15
C.3	Dimension arithmetic	15
C.4	Effect on conditioning	15
D	Exact Verification of the Certificate	15
D.1	What must be discharged	15
D.2	Rational constants and exact algebra	15
D.3	The verification script	15
D.4	What exact verification does and does not establish	15
E	The Extremal Control Law	15
F	The Moment-LP Dual and its Semidefinite Tightening	15
F.1	From the measure problem to its dual	15
F.2	Putinar's Positivstellensatz and the multipliers	15
F.3	Semidefinite assembly	15
F.4	Multi-phase assembly	15
F.5	Solver interface and verification of the certificate	15

G	Fixed-Grid Spectral Machinery	15
G.1	Chebyshev operators by direct recurrence	15
G.2	Ultraspherical banded operators	15
G.3	Conditioning	15
G.4	Boundary immobilization on a fixed grid	15
H	Sum-of-Squares in a Chebyshev Basis	15
H.1	Nondimensionalization and per-band scaling	15
H.2	Gram matrices in a non-monomial basis	15
H.3	Assembly and operator reuse	15
H.4	Conditioning claim	15

1 Introduction

1.1 Grid-based Hamilton-Jacobi-Isaacs evaluations

1.2 Fixed-grid pseudospectral methods

1.3 Thesis statement

1.4 Outline of contributions

2 The Model Problem

2.1 Abstract formulation of the multi-phase optimal control problem

The vehicle is a rigid body, not a point mass ([Appendix A](#)). Its state

$$x = (r, \theta, \phi, u, v, w, q_0, q_1, q_2, q_3, p, q, n_b) \in \mathcal{X} \subseteq \mathbb{R}^{13}, \quad (2.1)$$

augmented by mass, recession depth s and modal state $(\eta, \dot{\eta})$ where the corresponding mode or coupling is active, carries geocentric position, body-frame velocity, attitude quaternion and body rates as components. The control

$$c = (\vec{\delta}, \vec{\tau}_{\text{rcs}}, \varsigma) \in U, \quad (2.2)$$

collects flap deflections $\vec{\delta}$, reaction-control moments $\vec{\tau}_{\text{rcs}}$ and, in a propulsive mode, throttle ς ; $U \subset \mathbb{R}^{n_c}$ is compact.

Prove that U is compact.

2.2 The mode alphabet

2.3 Trajectory families as admissible words

2.4 The entry corridor: hypersonic path constraints

2.5 Certified envelopes for constitutive and coupling effects

2.6 Phase boundaries: physical, constitutive, and numerical

2.7 Asymptotic regimes and the dimensionless groups

2.8 Quantity of interest and uncertainty

3 Infinite-Dimensional Embedding and Measure Theory

3.1 Sobolev space formulation

3.2 Occupation measures and the weak Liouville identity

3.3 Per-mode measures, guard measures, and linkage

3.4 The embedding as a linear problem on a Banach space

4 Concentration-Compactness and Profile Decomposition

- 4.1 Prokhorov tightness as a preliminary
- 4.2 The singular limit and the scaling group
- 4.3 Tightness of the rescaled family
- 4.4 The dichotomy
- 4.5 The skeleton: concentration on the equilibrium glide manifold
- 4.6 The bubbles: boundary-layer profiles at each skip
- 4.7 The compact residual
- 4.8 Consequence: dimension reduction for the certificate

5 Bounding Limits and Certificates

5.1 Relaxation and the direction of inclusion

5.2 The comparison principle

5.3 Constructing the certificate from the decomposition

5.4 Verification over the corridor

5.5 Exact arithmetic and machine-checkable certification

5.6 Inner point cloud and the bounding gap

5.7 Limits of the analytical construction

6 Verification and Results

- 6.1 Validating the degenerate limit
- 6.2 Tightness against the extremal cloud
- 6.3 Independent numerical cross-check
- 6.4 The corridor constraints as hypotheses
- 6.5 Multi-mode words
- 6.6 Numerical evidence for the profile decomposition
- 6.7 Computational cost
- 6.8 Reproducibility environment

7 Discussion

7.1 Limitations

7.2 Scope: the boost-glide and aeroassisted class

7.3 Extensions

References

A Six-Degree-of-Freedom Equations of Motion

This appendix records the rigid-body dynamics the body of the paper treats as given. Nothing here is a contribution; the derivation is standard entry mechanics

A.1 State vector and frames

A.2 Translational equations

A.3 Rotational equations

A.4 Aerodynamic forces and moments

A.5 Atmosphere

A.6 Per-mode forms

A.7 Boundary discipline

B Corridor Constraints and Certified Envelopes

B.1 The corridor constraints in polynomial form

B.2 Compactness of the corridor

B.3 Aerodynamic coefficient envelopes

B.4 Ablation and aerothermoelastic envelopes

B.5 How the envelopes enter the certificate

C Polynomial Embedding of the Entry Dynamics

C.1 Augmentation

C.2 The augmented state and its constraint set

C.3 Dimension arithmetic

C.4 Effect on conditioning

D Exact Verification of the Certificate

D.1 What must be discharged

D.2 Rational constants and exact algebra

D.3 The verification script

D.4 What exact verification does and does not establish

E The Extremal Control Law

F The Moment-LP Dual and its Semidefinite Tightening

F.1 From the measure problem to its dual

F.2 Putinar's Positivstellensatz and the multipliers

F.3 Semidefinite assembly

F.4 Multi-phase assembly

F.5 Solver interface and verification of the certificate

G Fixed-Grid Spectral Machinery

G.1 Chebyshev operators by direct recurrence